

A note on a special measure

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April 13, 2020

Let Λ be the light cone

$$\Lambda = \{x \in \mathbb{R}^4 \mid x_4^2 = x_1^2 + x_2^2 + x_3^2\}.$$

This cone has many famous properties. One of them was found by Kowalski and Preiss [1]: the Lebesgue measure λ on this cone is uniform, that is, for any Euclidian ball $B_r(x)$ of radius r centered at $x \in \Lambda$, the value $\lambda(B_r(x))$ is independent of x . They found exact expressions for the values $\lambda(B_r(x))$. We will provide a different computation, which leads to the same result. First, for any Schwartz function Φ on $\mathbb{R}^4$,

$$\int_{\mathbb{R}^4} \Phi(x) d\lambda(x) = \sqrt{2} \iiint_{\mathbb{R}^3} \left(\Phi\left(x_1, x_2, x_3, \sqrt{x_1^2 + x_2^2 + x_3^2}\right) + \Phi\left(x_1, x_2, x_3, -\sqrt{x_1^2 + x_2^2 + x_3^2}\right) \right) dx_1 dx_2 dx_3 \quad (1)$$

since the gradient of the function $(x_1, x_2, x_3) \mapsto \sqrt{x_1^2 + x_2^2 + x_3^2}$ has norm one. Second, the uniformity of λ will follow if we prove that

$$\int_{\mathbb{R}^4} e^{-r|y_1 - x|^2} d\lambda(x) = \int_{\mathbb{R}^4} e^{-r|y_2 - x|^2} d\lambda(x) \quad \text{for any } y_1, y_2 \in \Lambda, r > 0. \quad (2)$$

We will prove that (2) leads to the uniformity at the end of the note. Since λ is homogeneous, it suffices to prove the assertion (2) for the case $r = 1$ only. Using the rotational symmetry of Λ , we reduce (2) to the case where $y_1 = cy_2$, where $c > 0$ is a constant. In other words (see formula (1)),

$$\iiint_{\mathbb{R}^3} \left(e^{-(x_1 - t)^2 - x_2^2 - x_3^2 - (\sqrt{x_1^2 + x_2^2 + x_3^2} - t)^2} + e^{-(x_1 - t)^2 - x_2^2 - x_3^2 - (\sqrt{x_1^2 + x_2^2 + x_3^2} + t)^2} \right) dx_1 dx_2 dx_3 \quad (3)$$

should be independent of $t > 0$. We are going to make the polar change of variables in this integral. Let us compute the part of the integral that does not depend on the radius $\sqrt{x_1^2 + x_2^2 + x_3^2}$ directly (this is, in fact, the computation of the Fourier transform of the Lebesgue measure on the sphere):

$$\begin{aligned} J_t(r) &= \iint_{x_1^2 + x_2^2 + x_3^2 = r^2} e^{2x_1 t} dx_1 dx_2 dx_3 = \sum_{\pm} \iint_{x_2^2 + x_3^2 \leq r^2} e^{\pm 2t\sqrt{r^2 - x_2^2 - x_3^2}} \frac{r dx_2 dx_3}{\sqrt{r^2 - x_2^2 - x_3^2}} = \\ &= \iint_{x_2^2 + x_3^2 \leq r^2} 2 \cosh\left(2t\sqrt{r^2 - x_2^2 - x_3^2}\right) \frac{r dx_2 dx_3}{\sqrt{r^2 - x_2^2 - x_3^2}} = 2\pi \int_0^r 2 \cosh\left(2t\sqrt{r^2 - \rho^2}\right) \frac{r \rho d\rho}{\sqrt{r^2 - \rho^2}} \stackrel{s=\sqrt{r^2 - \rho^2}}{=} \\ &= -2\pi \int_r^0 2r \cosh(2ts) ds = -4\pi r \left[\frac{\sinh(2ts)}{2t} \right]_{s=r}^{s=0} = \frac{2\pi r \sinh(2tr)}{t}. \end{aligned}$$

We return to the expression in formula (3) and use the Fubini Theorem (we do not need to multiply by the Jacobian since $J_t(r)$ is an integral over a 3-sphere of radius r , not 1 like in the classical polar change of variables):

$$\begin{aligned} \sum_{\pm} \iiint_{\mathbb{R}^3} e^{-(x_1-t)^2-x_2^2-x_3^2-(\sqrt{x_1^2+x_2^2+x_3^2}\pm t)^2} dx_1 dx_2 dx_3 = \\ \sum_{\pm} \int_0^\infty e^{-2r^2\pm 2rt-2t^2} J_t(r) dr = 2\pi \sum_{\pm} \int_0^\infty e^{-2r^2\pm 2rt-2t^2} \sinh(2tr) \frac{r}{t} dr. \end{aligned}$$

We have reduced the three-dimensional integral in (3) to a one-dimensional. We finish the computation:

$$\begin{aligned} 2\pi \sum_{\pm} \int_0^\infty e^{-2r^2\pm 2rt-2t^2} \sinh(2tr) \frac{r}{t} dr = \\ \pi \int_0^\infty \left(e^{-2r^2+4rt-2t^2} - e^{-2r^2-4rt-2t^2} \right) \frac{r}{t} dr = \pi \int_0^\infty \left(e^{-2(r-t)^2} - e^{-2(r+t)^2} \right) \frac{r}{t} dr = \\ \pi \left(\int_0^\infty e^{-2(r-t)^2} \frac{r-t}{t} dr + \int_0^\infty e^{-2(r-t)^2} dr - \int_0^\infty e^{-2(r+t)^2} \frac{r+t}{t} dr + \int_0^\infty e^{-2(r+t)^2} dr \right) \\ = \frac{\pi}{2t} - \frac{\pi}{2t} + \pi \int_0^\infty e^{-2(r-t)^2} dr + \pi \int_{-\infty}^0 e^{-2(r-t)^2} dr = \pi \int_{\mathbb{R}} e^{-2r^2} dr = \pi \sqrt{\frac{\pi}{2}}. \end{aligned}$$

Thus, (2) is proved. It remains to explain why this assertion leads to uniformity. Informally, the quantities $\lambda(B_r(x))$ are expressible in terms of the $\int e^{-r|x|^2} d\lambda(x)$ since the Gaussians are dense in any reasonable space of functions on the half-line. Here how the rigorous proof goes. It suffices to find (finite) collections of numbers $\{a_k\}_k$ and $\{s_k\}_k$ such that

$$\left| \lambda(B_r(x)) - \sum_k a_k \int e^{-s_k|y-x|^2} d\lambda(y) \right|$$

is arbitrarily small (and the collections $\{a_k\}_k$ and $\{s_k\}_k$ depend neither on x nor on λ). We estimate this difference by a suitable weighted L_2 -norm using the fact that $\frac{\partial}{\partial r} \lambda(B_r(x)) \lesssim (1+r)^3$ (this follows from (2) and dilation considerations):

$$\begin{aligned} \left| \lambda(B_r(x)) - \sum_k a_k \int e^{-s_k|y-x|^2} d\lambda(y) \right| = \left| \int_0^\infty (\chi_{[0,r)}(t) - \sum_k a_k e^{-s_k t^2}) d\lambda(B_t(x)) \right| \lesssim \\ \left| \int_0^\infty (\chi_{[0,r)}(t) - \sum_k a_k e^{-s_k t^2}) (1+t)^3 dt \right| \lesssim \int_0^\infty (\chi_{[0,r)}(t) - \sum_k a_k e^{-s_k t^2})^2 (1+t)^{10} dt. \end{aligned}$$

Thus, it suffices to show that the set of functions $\{e^{-st^2}\}_{s>0}$ is dense in the space $L^2(\mathbb{R}_+, (1+t)^{10})$ (the space of functions $f: \mathbb{R}_+ \rightarrow \mathbb{R}$ such that $\int f^2(t)(1+t)^{10} < \infty$). Assume the contrary, then there exists a non-zero function g in the dual space $L^2(\mathbb{R}_+, (1+t)^{-10})$ such that $\int g(t)e^{-st^2} dt = 0$ for any $s > 0$. Consequently, a non-zero tempered distribution h given by the formula

$$h(p) = \begin{cases} 0, & p < 0; \\ p^{-\frac{1}{2}} g(\sqrt{p}), & p > 0 \end{cases}$$

satisfies $\hat{h}(is) = 0$ whenever $s > 0$. The function $\hat{h}$ is analytic in the upper half-plane, so $\hat{h} = 0$ there. We also have $\hat{h}(\cdot, is) \rightarrow \hat{h}(\cdot, 0)$ as $s \rightarrow 0$ at least in $\mathcal{S}'(\mathbb{R})$. Therefore, $\hat{h} = 0$ on the real axis as a distribution, which contradicts the assumption that h is non-zero.

References

- [1] O. Kowalski, D. Preiss, *Besicovitch-type properties of measures and submanifolds*, Journal für die reine und angewandte Mathematik **379** (1987), 115–151.